Learning with Errors with Output Dependencies

LWE, LWR, and Physical Learning Problems under the Same Umbrella

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Is Learning With Errors (LWE) still secure if the error distribution is dependant of the exact value of the inner product?

Under which conditions? Is it useful?

Dimension 2: LPN vs. LPN-OD



Learning Parity with Noise (LPN_ε)

Given a secret $\boldsymbol{k} \in \mathbb{F}_2^n$, a sample $(\boldsymbol{a}, b)$ is generated as:

$\boldsymbol{a} \leftarrow \mathbb{F}_2^n$

$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e \pmod 2$

where $e \leftarrow \mathrm{Ber}(\varepsilon)$ *(Noise is independent of the key)*

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LPN with Output Dependencies ($\mathrm{LPN\text{-}OD}_{\varepsilon_0,\varepsilon_1}$)

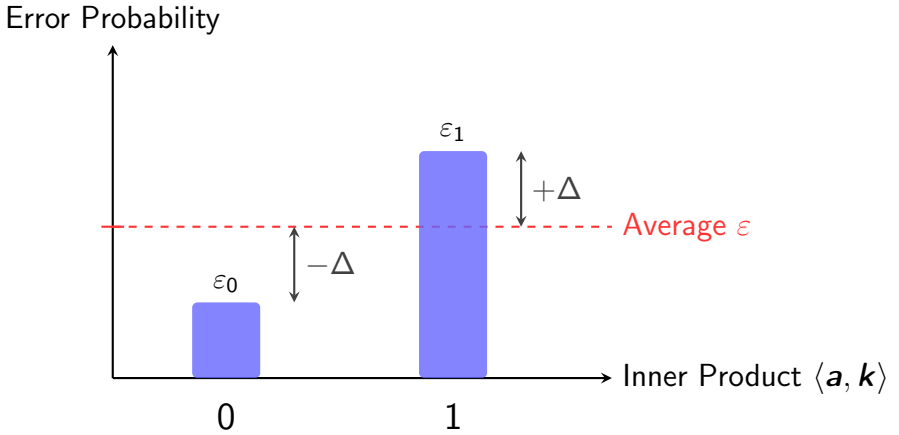
The error distribution now depends on $\langle \boldsymbol{a}, \boldsymbol{k} \rangle$:

$\boldsymbol{a} \leftarrow \mathbb{F}_2^n$

$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e \pmod 2$

where $e \leftarrow \mathrm{Ber}(\varepsilon_{\langle \boldsymbol{a}, \boldsymbol{k} \rangle})$ *(Noise depends on the internal value $\langle \boldsymbol{a}, \boldsymbol{k} \rangle$)*

Dimension 2 example



Each of the two Bernoulli is given by a single parameter, the probability that $e = 1$.

Dimension 2 solution



There exists a **tight** and **two-way** reduction from LPN-OD to standard LPN.[1]

Security Equivalence

The LPN-OD$_{\varepsilon-\Delta,\varepsilon+\Delta}$ problem is exactly as secure[a] as the standard LPN$_{\psi}$ problem, where the equivalent independent noise rate is:

$$\psi = \frac{\varepsilon - \Delta}{1 - 2\Delta}$$

[a] search problem variant

[1] Bellizia et al. – *Learning parity with physical noise: Imperfections, reductions and FPGA prototype* (CHES 2021).

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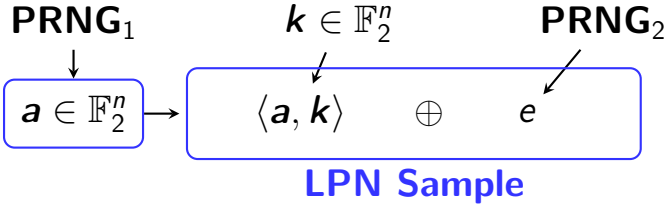
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But why is this a CHES paper?

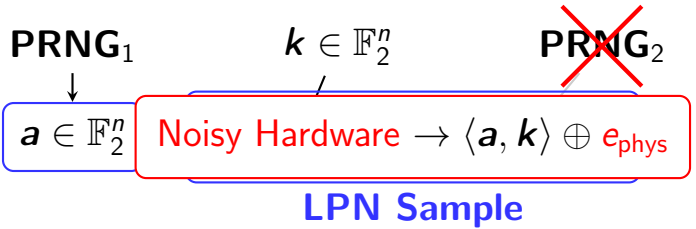
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First application: Inexact Computing



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Traditional Approach: Computing the error e requires an expensive secure PRNG.
Inexact Computing: Hardware imperfections (eg. over-voltage) natively fault inner product, guaranteeing masking friendly and side-channel friendly implementation. Impossible to totally guarantee the error distribution dependancies.

The Follow-up (Bellizia et al., CHES 2022)

*Extending this concept to larger moduli by designing usable instances of Learning With Physical Errors (***LWPE***).*

Generalizing to Dimension q: LWE vs. LWE-OD



Learning With Errors (LWE_{ψ})

Given a secret $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$, a sample $(\boldsymbol{a}, b)$ is generated as:

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LWE with Output Dependencies ($\mathrm{LWE\text{-}OD}_{\varphi_0,\ldots,\varphi_{q-1}}$)

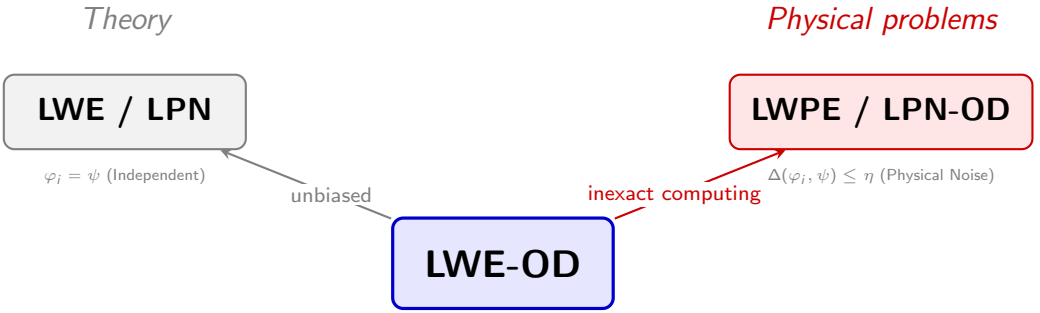
The error distribution is now a **family** of distributions:

$\boldsymbol{a} \leftarrow \mathbb{Z}_q^n$

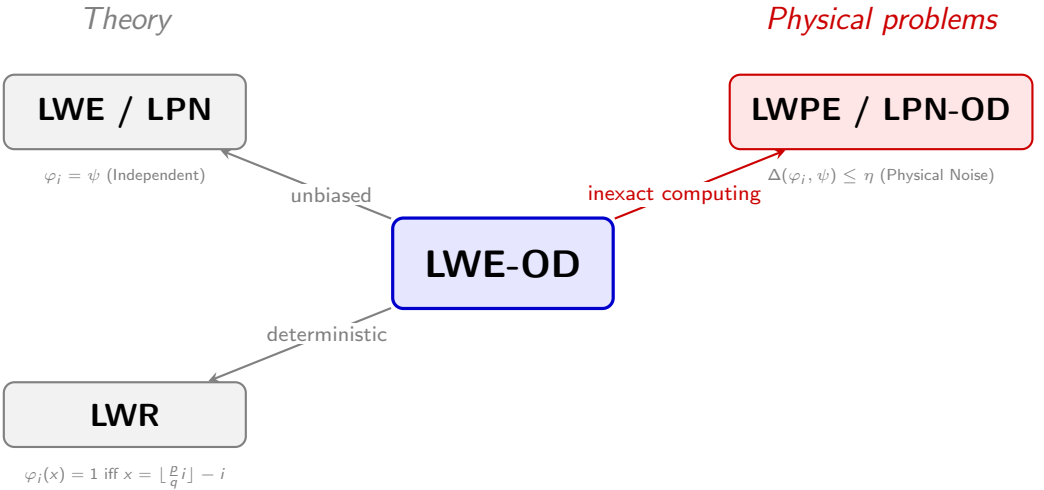
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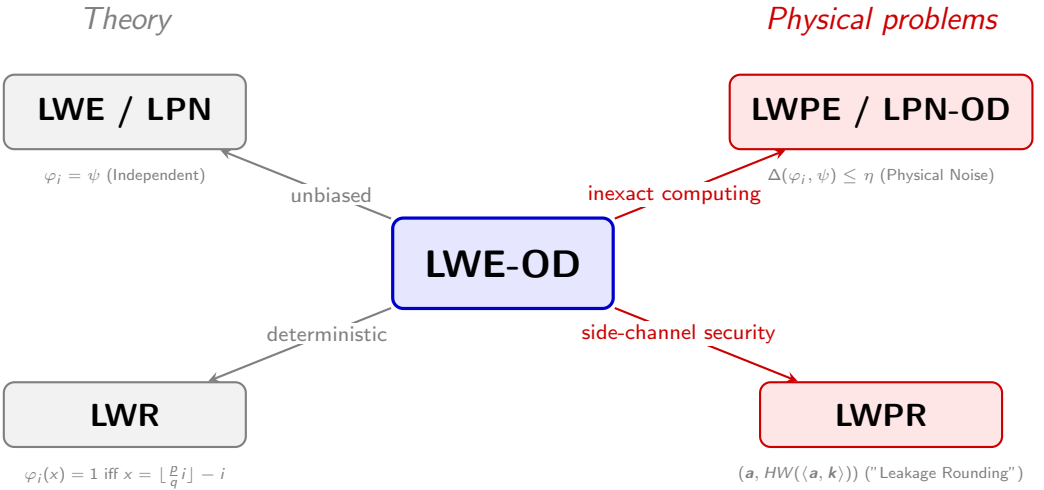
Applications: A Unified Framework



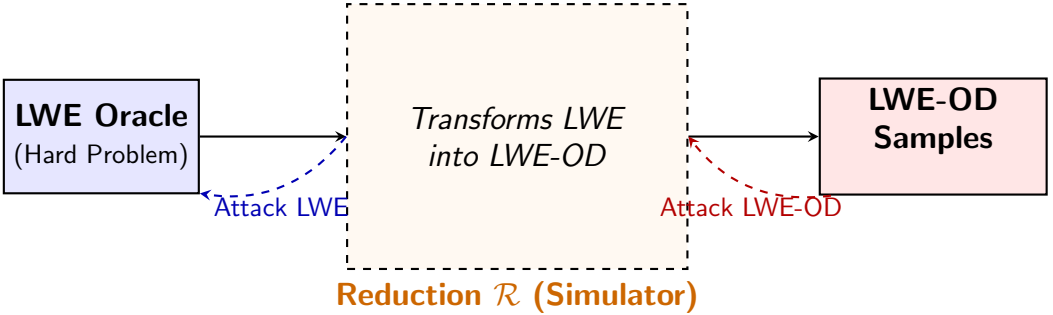
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Reduction Strategy



Step 1: Characterizing LWE-OD Samples



The last coefficient is tilted:
The marginal distribution of b is no longer uniform:

$$\rho(j) = \Pr[b = j] = \frac{1}{q} \sum_{k=0}^{q-1} \varphi_k(j - k)$$

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Equivalent characterization:

$$(\boldsymbol{a}, b) \leftarrow \text{LWE-OD}_{\varphi_0,\ldots,\varphi_{q-1}} \iff \begin{cases} \Pr[b = j] = \rho(j) \\ \Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] = \frac{1}{q^{n-1}} \cdot \varphi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \rho(j) \end{cases}$$

Step 1: Towards a Rejection Sampler



Input: A tilted distribution ρ, a family of LWE_{χ_j} samplers
Output: A sample $(\boldsymbol{a}, b_1)$
$b_0 \leftarrow \rho$;
do
  $(\boldsymbol{a}, b_1) \leftarrow \text{Sample-LWE}_{\chi_{b_0}}$;
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We need on-demand access to q **different LWE oracles** simultaneously with (potentially) different error distribution. This naturally motivates the introduction of an intermediate problem.

Step 1: Intermediate Problem ($\mathrm{qLWE}_{\chi_0,\ldots,\chi_{q-1}}$)



Definition: $\mathrm{qLWE}_{\chi_0,\ldots,\chi_{q-1}}$

A batch of q independent LWE samples with a shared secret $\boldsymbol{k}$ and **distinct error distributions** χ_i.

$$\text{qLWE sample} := ((\boldsymbol{a}_0, b_0), \ldots, (\boldsymbol{a}_{q-1}, b_{q-1}))$$

where $b_i = \langle \boldsymbol{a}_i, \boldsymbol{k} \rangle + e_i \pmod q$ and $e_i \leftarrow \chi_i$.

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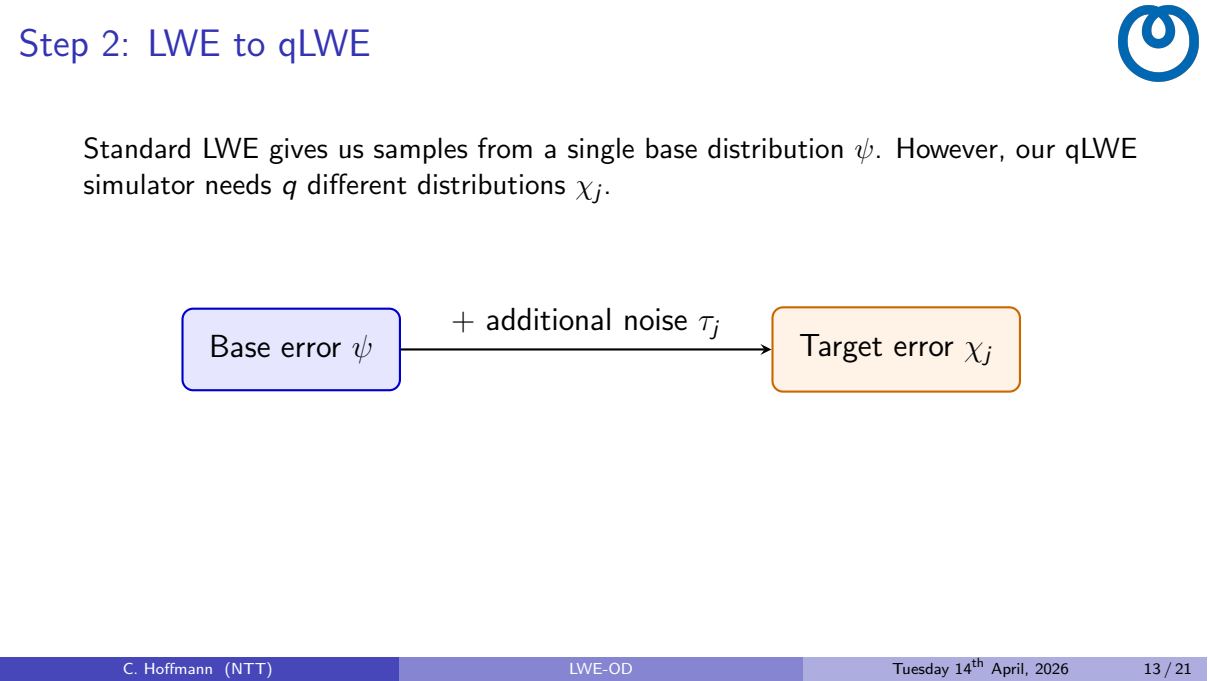
where $b_i = \langle \boldsymbol{a}_i, \boldsymbol{k} \rangle + e_i \pmod q$ and $e_i \leftarrow \chi_i$.

Theorem 1: Reduction from qLWE to LWE-OD

$\text{LWE-OD}_{\varphi_0,\ldots,\varphi_{q-1}}$ is **at least as hard** as $\text{qLWE}_{\chi_0,\ldots,\chi_{q-1}}$ iff:

$$\chi_j(i) = \frac{\varphi_{j-i}(i)}{q \cdot \rho(j)} \quad \text{where} \quad \rho(i) = \frac{1}{q} \sum_{k=0}^{q-1} \varphi_k(i-k)$$

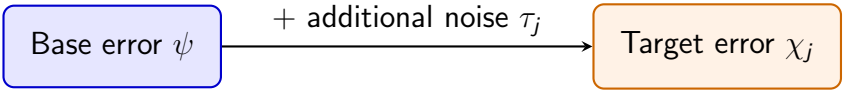
Step 2: LWE to qLWE



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We derive every equation from the base one. Adding independent errors results in the convolution of their distributions:

$$\chi_j = \psi * \tau_j$$

Step 2: Inverting the Convolution



The Walsh-Fourier Transform

For a distribution χ over $\mathbb{Z}_q$, its transform is defined as:

$$\widehat{\chi}(y) = \sum_{x \in \mathbb{Z}_q} \chi(x)\omega^{-xy} \quad \text{where } \omega = e^{2i\pi/q}$$

Convolution in becomes point-wise multiplication in the Walsh domain:

$$\widehat{\chi_j} = \widehat{\psi} \cdot \widehat{\tau_j} \quad \Longrightarrow \quad \widehat{\tau_j} = \frac{\widehat{\chi_j}}{\widehat{\psi}}$$

Step 2: Theorem



Input: Base oracle LWE_{χ_0}, error distributions τ_j
Output: Target sample for qLWE_{χ_j}
$(\boldsymbol{a}, b) \leftarrow \text{Sample-}\mathsf{LWE}_{\psi}$;
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Theorem 2: Reduction from LWE to qLWE

The reduction is valid if the additional noise τ_j is a valid probability distribution: $\mathsf{qLWE}_{\chi_0,\ldots,\chi_{q-1}}$ is at least as hard as LWE_{ψ} iff

$$\tau_j(x) = \mathcal{W}^{-1}\left(\frac{\widehat{\chi_j}}{\widehat{\psi}}\right)(x) \in \mathbb{R}_{\geq 0}$$

Theorem 3: The Full Reduction



Theorem 3: LWE to LWE-OD Reduction

LWE-OD$_{\varphi_0,\ldots,\varphi_{q-1}}$ is **at least as hard** as LWE$_\psi$ if:

$$\forall i, j \in \mathbb{Z}_q \text{ such that } \rho(i) \neq 0 :$$

$$\sum_{k \in \mathbb{Z}_q} \frac{1}{\hat{\psi}(k)} \sum_{l=0}^{q-1} \varphi_{i-l}(l) \omega^{k(l-j)} \in \mathbb{R}_{\geq 0}$$

Direct composition of the first two theorems

Provides an equation to verify if there is a security equivalence.

Limitations: The Mapping Challenge



Theorem 1: $\textbf{LWE-OD}_{\varphi_0,\ldots,\varphi_{q-1}} \rightarrow \textbf{qLWE}_{\chi_0,\ldots,\chi_{q-1}}$

Direct: The family (χ_i) is always uniquely derived from (φ_i).

Inverse: Non-trivial to identify which (φ_i) corresponds to a given (χ_i).

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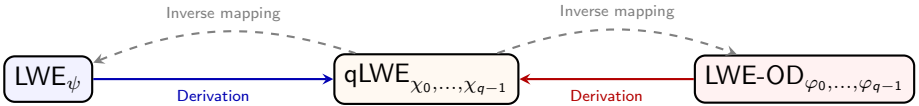
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Theorem 3: No generic correspondence



Sanity Checks



Standard LWE
If $\varphi_i = \psi$, the reduction becomes an identity mapping.

LPN-OD ($q = 2$)
The roots of unity are always in $\mathbb{R}$, our general reduction gives the same result as the state-of-the-art.

LWR reduction



LWR as an LWE-OD problem

For an input $i \in \mathbb{Z}_q$, the deterministic rounding error is $f(i) = \left\lfloor \frac{p}{q} i \right\rceil \cdot \frac{q}{p} - i \pmod{q}$.
Therefore, $\mathsf{LWR}_p \equiv \mathsf{LWE\text{-}OD}_{\varphi_0,\ldots,\varphi_{q-1}}$ with Dirac error distributions: $\varphi_i = \delta_{f(i)}$

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Case $p \mid q$

When p divides q, the error residues $f(i)$ are perfectly periodic. This structure allows for the (already known[a]) reduction:

$$\mathsf{LWR}_{q,p} \text{ is } \textbf{at least as hard} \text{ as } \mathsf{LWE}_{q,\psi}$$

$$\text{where } \psi = U_{\{0,\ldots,q/p-1\}}$$

[a] Bogdanov et al., On the hardness of learning with rounding over small modulus.

General Case: $p \nmid q$



Conjecture 1: "More noise is more secure"

For $a < 0$, $\mathsf{LWR}^n_{p,q}$ with $\psi = \mathcal{U}([a, 0])$ is **at least as hard** as LWE^n_{ψ} with $\psi = \mathcal{U}([a + 1, 0])$.

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Theorem 4: General Hardness of LWR

Assuming Conjecture 1, for any $q > p \geq 2$ and $L = \lfloor q/p \rfloor$:

$\mathsf{LWR}_{q,p}$ is **at least as hard** as $\mathsf{LWE}_{q,\psi}$

with $\psi = \mathcal{U}(] - L + 1, 0])$.

Conclusion and Future Work



Takeaway

LWE-OD provides a **unified umbrella** for analyzing theoretical (LWE, LWR) and physical learning problems (LPN-OD, LWPE) within a single reduction framework. There exists a first reduction that allows for some applications.

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Research directions

Is it possible to use a completely different reduction, at least for the second step? Is there a way to approximate the physical problems (eg. noisy variants) to make them fit in this result?